

Centralized Decision Making and Informed Lobbying[†]

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We address the trade-off between centralized and decentralized decision making subject to influence from privately informed lobbies. We focus on informative equilibria with separating differentiable contribution schedules and identify an information transmission effect under centralized structures. Such effect decreases capture and increases welfare when lobbies have “aligned preferences.” The opposite effect holds for “polarized preferences.” We present two examples of this framework: local public goods and customs union agreements. Finally, we compare the policy outcomes from this political economy perspective to those under a normative mechanism design approach, and extend our analysis to the discussion of pooling equilibria. (JEL D72, D82, F13, F15, H41)

A classical question in public economics relates to the costs and benefits of centralization versus decentralization of public decision making, and the allocation of decision rights inside governments. In his seminal work on the provision of local public goods, Oates (1972) addressed this issue, emphasizing on one hand the trade-off between the internalization of cross-district externalities (favoring centralized systems), and on the other hand, the reactivity to local citizen preferences (favoring decentralized systems). A subsequent literature extended that discussion to a political economy perspective, recognizing that centralized and decentralized public systems face different political constraints and incentives in terms of public spending and taxation (Inman and Rubinfeld 1997, Lockwood 2002, and Besley and Coate 2003).

An important dimension along this line of research has been to highlight how different levels of governments may be subject to political influence by special interest groups that compete with each other in order to affect the policy process (Bardhan and Mookherjee 2000). Two main channels of influence have been thereof emphasized. The first one relates to the presence of private information that can be

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disclosed and strategically distorted by specific groups in order to obtain policies better fitting their own agenda. Alternatively, interest groups may also influence policymakers through bribes or financial contributions. In such a context, a number of natural questions arise. How do these channels of influence interact with the structure of decision making inside the government? How do these interactions frame the degree of competition for political influence and what are the consequences for the equilibrium public policies? What are the implications for the analysis of the costs and benefits of centralized and decentralized systems? What should be the optimal structure of government from the point of view of society?

The purpose of this paper is to consider these issues in a simplified context in which the two sources of lobbies' influence—information transmission and contributions—are closely connected. Specifically, we consider situations in which special interest groups have private information on how policies affect their payoffs, and use financial contributions to influence politicians and policymakers. In such a setting, we compare the influence of informed lobbying in centralized and decentralized public decision structures. Our departing point of analysis is to recognize two well-known but important features of centralized systems: (a) they concentrate the political competition for influence at a higher hierarchical and more encompassing level of government than decentralized systems; (b) they also tend to generate more policy uniformity across localities than decentralized structures.

Starting first with the case with perfect information, we show that feature (a) tends to reduce the degree of political capture in centralized systems. Interestingly, the precise mechanism through which that works depends on the structure of local lobbies' preferences for policies. Specifically, it depends on whether lobbies' ideal policies lie on the same side of the policy spectrum (the "aligned preferences" case) or on opposite sides (the "polarized preferences" case). In a context with aligned preferences, lobbies wish to shift policies in the same direction, even though they might disagree on the intensity of the shift. In such a case, centralization then weakens the competition for influence through what can be called a *preference dilution* effect. As a consequence, centralized policies tend to accommodate more preference trade-offs across locations. The scope of influence by lobby groups located in different areas is "diluted" compared to decentralized systems. This *preference dilution* effect in turn pushes for policies that are more aligned with the general public interest.

Conversely, in a context with polarized preferences, lobbies want to shift policies in opposite directions. In this case, centralization induces fierce competition since one lobby must overcome her¹ rival's influence to shift the policy in her favor. This strong "Bertrand" type of *competition* effect also tends to reduce capture as lobbies tend to neutralize each other. Only the strongest lobby exerts effective influence, and only to the extent that her strength outweighs her rival's.

Our main contribution is to show that in a context of asymmetric information, centralization also induces an *information transmission* effect that interacts with the competition between lobbies. This effect arises in centralized systems because

¹ We refer to lobbies with feminine pronouns and to policymakers with masculine pronouns.

policies integrate cross-district specificities and therefore create strategic information interdependencies for lobbies located in these districts. In fact, a centralized policymaker's willingness to grant a favorable policy to a lobby located in one area depends on how much that policy is also serving a rival lobby in another area. The existence of lobby-specific private information implies that each interest group's optimal influence contribution depends not only on her own private information but also on the private information possessed by the rival group. Through this channel, strategic information transmission crucially interacts with political influence (i.e., bribes and financial transfers).

Under centralization, each lobby proposes financial contributions that may reveal part of her private information to the common policymaker. In equilibrium, this feature allows the common policymaker to learn something about each lobby's private information. Each lobby finds this piece of information relevant for the design of her optimal influence strategy, and has incentives to screen out from the policymaker what the latter learned from rival lobbies. Screening, however, is costly and induces lobbies to exert less influence. Conversely, screening increases the common policymaker's bargaining power, and provides him political (informational) rents.

Again, we show that the impact of the *information transmission* effect for the degree of competition between lobbies depends on the preference structures of these lobbies. In the "aligned preferences" case, the *information transmission* effect reduces political capture of centralized decisions. As said, this results from the fact that screening the policymaker's acquired information leads to higher costs in terms of financial lobbying contributions. As a consequence, lobbies use less of them and therefore exert less influence. With polarized preferences, the results are reversed. Again, screening distorts financial contributions. This effect, however, is now asymmetric and more intense for weaker lobbies. As a consequence, strong lobbies end up exerting more influence, leading to an overall increase in political capture in centralized policies.

Our third contribution is to discuss the conditions under which centralized systems are preferable to decentralized ones and the role that information transmission plays in such trade-offs. With aligned preferences, the trade-off weights the *preference dilution* effect and the *information transmission* effect against the standard costs of making a uniform central policy (feature (b) of centralized systems). Specifically, our analysis indicates that the larger the stake of lobbies' private information, the stronger the *information transmission* effect and the *dilution* effect and, therefore, the more likely the centralized regime dominates the decentralized regime from a normative perspective. Conversely, in the setting with polarized preferences, the choice between centralization and decentralization compares the *competition* effect (in favor of centralization) against the policy *uniformity* and *information transmission* effects (in favor of decentralization). As a consequence, a higher degree of information asymmetry makes it more likely that decentralized structures dominate centralized structures.

We illustrate our model's implications in two classical problems of joint policymaking: the provision of local public goods and the setting of an import tariff in customs union agreements. The first application is quite direct: the amount of public good is a local policy decision. This decision can be undertaken by a specific

policymaker for each district (decentralized structure), or by a common policymaker for both districts (centralized structure). In the centralized case, we differentiate the system on the policy space (uniform versus nonuniform policy) and on the financing mode of the public good (separate or common pool budget). This distinction leads to different policy preference structures for lobbies. More precisely, a uniform policy and separated district budgets induce lobbies to have aligned preferences. As a consequence, lobbying competition is weak and information transmission under centralized structures reduces political capture. Conversely, with nonuniform policies and a common pool budget, lobbies' preferences get polarized, and competition is fierce. Information transmission in this case hinders competition and consequently increases political capture.

The second application studies how private information affects political incentives in a customs union trade agreement. In this agreement, countries remove all import tariffs between them and set uniform trade policies toward countries outside the union. This type of agreement, therefore, corresponds to a centralized decision structure. Conversely, when countries do not sign an agreement, the choice of the import tariff is decentralized. Protectionist lobbies representing similar import competing industries from different countries have aligned preferences toward tariff protection. As a consequence, the information transmission effects make it more likely to have a welfare improving customs union agreement.

Finally, the last part of the paper discusses two important extensions of our analysis. First, we compare our standard political economy approach (PEA) in which lobbies act as informed principals, implicitly committing to financial contributions in order to influence a uniformed agent (the policymaker) to a mechanism design approach (MDA) in which the hierarchy of moves is inverted, and the policymaker acts as the uniformed principal proposing a policy and transfers schedules to lobbies that now behave as informed agents. Under decentralization, the MDA leads in each district to a classical principal-agent relationship; while under centralization, the MDA leads to a uniformed principal and two privately informed lobbies under the restriction of a uniform policy schedule implemented across districts. In both environments, the principal faces the typical trade-off between socially favorable policies and informational rents left to the lobbies. The equilibrium solutions of the MDA under decentralization and centralization allows a comparison between the outcomes reached under MDA and those obtained under PEA. As it turns out, the structure of lobbies' preferences again tends to matter for such comparison. In particular, under MDA and aligned lobbies' preferences, the equilibrium policies under decentralization and centralization get closer to the welfare optimum, and aggregate welfare increases for both regimes. Because of the competitive effect between lobbies, the opposite result, however, holds for the centralization regime with polarized preferences.

Second, we complete our analysis by an examination of pooling equilibria in our lobbying game under centralization. We extend our main discussion focusing on separating (informative) equilibria of the lobbying game under centralization to other non-informative equilibria that may also exist in our game. Specifically, we construct pooling equilibria and discuss the conditions for their existence and implications by comparing with the separating equilibrium.

The plan of the paper is the following. The next subsection discusses the related literature. Section I introduces our basic linear quadratic model of policymaking under lobbying influence with two social entities and one lobby group associated with each entity. It also computes the benchmark policies of the political game under centralization and decentralization under perfect information. Section II considers the case with lobby-specific private information. In particular, we provide the fully explicit characterization of the equilibrium policies and contributions schedules. Section III discusses the optimality of the centralized and the decentralized structures under both perfect and private information. Section IV provides the application of our simple parametric example to the contexts of local public good provision and customs union agreements. Section V considers the policy game under the mechanism design approach (MDA), and discusses as well the implications of pooling equilibria in the political economy lobbying game. Finally, Section VI concludes with avenues for future research.

Related Literature.—This paper investigates how capture affects policy decisions according to the structure of public decision making. It is closely related to Bardhan and Mookherjee (2000) and Bordignon, Colombo, and Galmarini (2008) which have also approached this problem. In Bardhan and Mookherjee (2000), centralization is better when lobbies are less well organized at the national level, while decentralization dominates when local districts have a strong preference for one party. Centralized and decentralized structures have different impacts because voters have different levels of awareness and lobbies have different levels of cohesion in the two-decision structures. In our case, the centralized and decentralized structures affect political competition differently because of information screening and its implications for lobbying incentives.

Bordignon, Colombo, and Galmarini (2008) consider lobbying under centralization and decentralization in a setting with interdistrict externalities. They find that (de)centralization is better when the lobbies' preferences are (not) aligned. Our paper does not consider interdistrict externalities. Mostly, we show that the *information transmission* effect we identify can reverse their results making centralized decision making eventually less desirable.

Our paper is related to the classical work of Oates (1972). As in Oates (1972), we allow for heterogeneity in districts' preferences. However, to focus on the effects of lobbying and *information transmission* in the simplest possible way, we do not consider interdistrict spillovers. Hence, in our setting, decentralization is always welfare superior in the absence of lobbying. The benefits from centralization come uniquely from hindering lobbying influence through *preference dilution* and *information transmission* in the "aligned preferences" setting or through the *competition* effect in the "polarized preferences" setting. As is well known in a perfect information context, introducing district spillovers would make the case for centralization even stronger.

In the context of local public good provision, our paper connects to several papers addressing different political economy aspects of the trade-off between centralization and decentralization (Seabright 1996, Lockwood 2002, Redoano and Scharf 2004), but without asymmetrically informed lobbying influencing the process. Related to our second application to custom unions, we also connect to a large literature on

the political economy of trade agreements (de Melo, Panagariya, and Rodrik 1993; Richardson 1993; Grossman and Helpman 1994; Krishna 1998; Ornelas 2005a,b, and Maggi and Rodríguez-Clare 2007). While like us, de Melo, Panagariya, and Rodrik (1993) identify a preference dilution effect of lobbying under a trade agreement, in contrast to us, this literature does not consider trade lobbying with asymmetric information.

Our paper also connects to the large political science literature such as Austen-Smith (1995), Austen-Smith and Wright (1992), Potters and van Winden (1992), and Bennedsen and Feldmann (2006) that consider lobbies as potentially useful providers of information to policymakers. Our work follows a different approach, closer to Lima and Moreira (2014) who treat lobbies as rent-seekers with private information about their own preferences or technologies.

From a technical perspective, our analysis borrows from the literature on informed principal problems (Maskin and Tirole 1990), and the recent theoretical literature on common agency with privately informed principals such as Martimort and Moreira (2010) and Lima and Moreira (2014). The first literature provides the appropriate framework to analyze our political game under decentralization, while the second allows us to characterize the political game under centralization. We apply the techniques developed therein to contrast how centralized versus decentralized policy structures differently affect political competition between privately informed interest groups. Additionally, our work provides the new qualitative result that for the “polarized preferences” case, information transmission drives policies away from the policymaker’s bliss point. This result contrasts with Martimort and Semenov (2008). They also analyze a common agency lobby game with polarized preferences for the lobbies (principals) but with private information on the agent’s (government) preferences. They find that private information on the government side reduces the lobbies’ influence. Hence, our qualitative result for the “polarized preferences” setting is new in this literature. It uncovers precisely when the *information transmission* effect may contribute to political capture in centralized systems.

I. The Model

We consider an economy with two distinct entities (groups, districts, communities, countries, ...), A and B . In each entity $i \in \{A, B\}$, a policymaker is needed to implement a local policy² p_i . Each entity is composed of two types of individuals with different preferences regarding the implementation of policy p_i . There is a continuum of identical individuals (with normalized mass of 1) in entity i having the following preferences:

$$(1) \quad W_i(p_i) = -\frac{1}{2}(p_i - \alpha_i)^2,$$

where α_i reflects the individual’s preferred policy level in entity i .

²For example, policy p_i can be the amount of a local public good, a specific local tax, or a regulation when the entities are geographic districts within the same national territory. It can be a “border” policy such as trade, immigration, or international capital flow regulations when the entities themselves are national governments.

There is also a politically organized lobby i . That lobby reflects the interests of a small fraction of agents in entity i that have different preferences from the individuals in i . On top of that, the lobby can disburse money to influence the policymaker responsible for the implementation of the policy. More precisely, we assume that the lobby's objective function can be described as

$$(2) \quad V_i(\theta_i, p_i, C_i) = -\frac{1}{2}(p_i - \theta_i)^2 - C_i,$$

where θ_i is the lobby's policy bliss point (preferred policy) and C_i the amount of money contributions to influence policymaking. Generically, the two lobbies and the society have distinct bliss points. We will focus on two main settings for the configuration of these bliss points. In the aligned setting, the lobbies' bliss points are to the right side of the society's average policy $\hat{p}$ (that is, $\theta_A, \theta_B > \hat{p}$). Lobbies may disagree on their ideal level, but both want to push policies away from the society's average ideal policy.

In the polarized setting, lobby A's bliss point is to the right of $\hat{p}$, while lobby B's bliss point is to the left of $\hat{p}$ (that is, $\theta_B < \hat{p} < \theta_A$). In that case, the lobbies' desire is to push policies in different directions away from the society's average ideal policy.

Under decentralized decision making, each entity i is endowed with one policymaker. As is common in the influence lobbying literature (Bernheim and Whinston 1986 and Grossman and Helpman 1994, 1995), we assume that this policymaker cares about the society's welfare, W_i , but also likes money contributions, C_i . His preferences are given by

$$U_i(p_i, C_i) = C_i + \lambda W_i(p_i),$$

where λ is the relative preference between contributions and the society's welfare function.

Under centralization the two entities can delegate the policy decision to a joint policymaker. In that case, this agent cares about the aggregate society's welfare and can be influenced by both lobbies. Additionally, we assume that the policymaker has to set a common policy p for both entities. This assumption of policy uniformity is natural when, by definition, centralization imposes a common policy instrument between the two entities. For instance, this is the case with a custom union or a regional economic union that uniformly regulates "border" policies of different national entities. In the case of fiscal federalism, this feature is not necessarily satisfied and may demand specific assumptions (see Besley and Coate 2003, Lockwood 2002, and Loeper 2011). Still, as a first pass, it may be useful to capture the idea that centralized decision making is less sensitive to local specificities than decentralized decision making. Moreover, as will be clear in the sequel, this assumption is not crucial for our conclusions. What is important for the *information transmission* effect that we identify is that the policy of one entity generates externalities (any kind of externality) for the other entity. In this respect, uniformization of policies induces a public good component of centralized

policymaking, which is then the simplest case of externalities that we need for our political game.

Specifically, the preferences of the joint policymaker under centralization can be represented as

$$(3) \quad U(p, C_A, C_B) = \sum_i [C_i + \lambda W_i(p)],$$

i.e., the sum of the utilities of the decentralized case over the two entities.

The timing of the game is as follows:

- (i) In each entity $i \in \{A, B\}$ nature draws the lobby types θ_i ;
- (ii) Lobbies offer contributions to the policymaker(s);
- (iii) Policymakers accept or reject the contributions;
- (iv) Policies are set; if contributions are accepted, payments are made accordingly.

Note that with our specification of preferences for policies (1), (2), and (3), we assume that the parameters θ_i characterizing the preferences of lobbies (2) do not directly affect the policymaker's community preferences $W_i(p)$, which depend only on p . This is somewhat in contrast with the Grossman and Helpman (2001) approach to lobbying by which lobbyists tend to act on behalf of some elements of the local communities, and hence θ_i would also appear directly into the policymaker objective function. Having private information of lobbyists in the preferences of the policymaker brings additional complex signaling issues in the multi-informed principals/agent game.³ Our preferences specification allows us to isolate neatly how screening considerations already generate interesting information effects associated to centralized structures.

A. Benchmarks

To understand the effects of lobbying and political influence, it is useful to present first the benchmark results for decision making under perfect information.

Decentralization.—With decentralized policies and no political influence, each policymaker chooses the policy that maximizes the society's preferences. In our simple setup, this is exactly the society's preferred policy α_i . Therefore, the optimal decentralized policy of district i is $\check{p}_i = \alpha_i$ and the social welfare is given by $W_i(\check{p}_i) = 0$, for all i .

Another important benchmark is the decentralized decision making under lobby influence and perfect information. Following the long literature on rent-seeking, we approach this problem assuming the lobby offers contributions to a influenceable

³Typically, there will be signaling to multiple audiences: in equilibrium, the contribution schedule of lobby i provides a signal not only to the policymaker (the agent) but also to lobbyist j .

policymaker. This political game of influence collapses to a simple principal-agent model where each lobby incentivizes her local policymaker to implement her favored policy p_i . More precisely, given the realization of her specific parameter θ_i , the lobby of each entity i solves the following program:

$$\max_p -\frac{1}{2}(p - \theta_i)^2 - C_i,$$

subject to the policymaker's participation constraint

$$C_i - \frac{\lambda}{2}(p - \alpha_i)^2 \geq 0.$$

The policy that solves this problem is given by

$$(4) \quad \tilde{p}_i(\theta_i) = \frac{\theta_i + \lambda \alpha_i}{1 + \lambda},$$

i.e., the optimal policy under political influence is a weighted average between the lobby's and the society's bliss points. In decentralized decisions, there are no interdependencies between entities. As a result, the policy is not a function of the other entity's characteristics.

Centralization.—When the policymaker has to set a uniform policy for both entities and there is no political influence, he solves the following program:

$$\max_p -\frac{1}{2} \sum_i (p - \alpha_i)^2,$$

which has the following optimal policy:

$$\hat{p} = \frac{\alpha_A + \alpha_B}{2},$$

which is simply the average of the districts' optimal policies. Social welfare under centralization provides $W_A(\hat{p}) + W_B(\hat{p}) = -(\alpha_A - \alpha_B)^{2/4} < 0$. Obviously this negative value reflects the loss from policy uniformization. We refer to it as the *uniformization* effect. Its magnitude depends on the distance between the entity's bliss points. Without political influence, decentralization yields higher payoffs since policies are tailored to meet the entities' social preferences. Under centralization, neither entity gets its preferred policy. By construction, the model has a “decentralization bias,” since we do not introduce any cross-entity externality, which is the usual driving force for policy centralization.

Let us turn to the last benchmark case of political influence. In a centralized structure, the policy is common to both entities. As a consequence, there is a public good component for both lobbies who offer contributions to the common policymaker. While the policymaker now cares about the welfare of both districts, he is also subject to the influence of the two lobbies. After the realization of the specific parameters θ_A and θ_B , the political game becomes a standard common agency game in which each lobby i proposes a contribution schedule $C(p, \theta_i)$ to influence the choice

of p . We follow Bernheim and Whinston (1986) and, as usual, assume that lobbies play truthful strategies. Thus, the equilibrium of the political game is equivalent to the solution of the following centralized program:

$$\max_p -\frac{1}{2} \sum_i (p - \theta_i)^2 - \frac{\lambda}{2} \sum_i (p - \alpha_i)^2,$$

whose first-order condition is

$$(5) \quad -[(p - \theta_A) + (p - \theta_B)] - \lambda[(p - \alpha_A) + (p - \alpha_B)] = 0.$$

The policy that solves this necessary and sufficient first-order condition is

$$(6) \quad \bar{p}(\theta_A, \theta_B) = \frac{\theta_A + \theta_B + \lambda(\alpha_A + \alpha_B)}{2(1 + \lambda)}.$$

Notice that as $\theta_A + \theta_B \rightarrow \alpha_A + \alpha_B$, the policy tends to the welfare optimal uniform policy $\hat{p}$.

For the aligned setting, $\theta_A, \theta_B > \hat{p}$ and the equilibrium policy is greater than the welfare maximizing policy under centralization. Equation (6) shows that, under centralized decision making, the equilibrium policy reflects both the society's average preference and the lobbies' preferences.

When preferences are polarized, we have that $\theta_A > \hat{p} > \theta_B$ and the first term of $\bar{p}(\theta_A, \theta_B)$ can be higher or lower than $\hat{p}$, depending on the sign of $\theta_A + \theta_B$. This policy can, therefore, be above or below the welfare maximizing policy. Another way to understand the differences between the polarized and aligned settings is to make a variable transformation so that both θ s are greater than $\hat{p}$, and that lobby B 's preferences are given by $-(p + \theta_B)^2/2$. This implies that lobby B 's bliss point is $-\theta_B$. With this transformation, the equilibrium policy under the polarized setting writes as

$$(7) \quad \bar{p}(\theta_A, \theta_B) = \frac{\theta_A - \theta_B + \lambda(\alpha_A + \alpha_B)}{2(1 + \lambda)}.$$

From now on, we will assume that this transformation holds. Notice that only the lobby with the greatest θ exerts influence, and only up to the amount of the difference between the two bliss points. If $\theta_A - \theta_B = \alpha_A + \alpha_B$, the centralized policy again is equal to the welfare maximizing policy.

II. Lobbying with Private Information

Consider now the situation where policymakers can be influenced by privately informed interest groups. We model the lobbies as the principals of an agency game. They offer contributions in order to obtain a favorable policy. The lobbies also have private information about their ability to exert influence. We focus on the interplay between lobbying and information asymmetry under centralized and decentralized structures.

Lobbies are privately informed about the parameter θ . As a result, the influence level is unknown *ex ante* by the society and by the policymaker, and does not have any direct effect on the society's welfare. To keep the model simple, this piece of information is not directly policy relevant. The intuition is that the content of private information is linked to the structure of organizational costs of the lobby or the preferences of the sectors the lobbies represent. As such, we assume that it has a negligible weight on the society's welfare. This obviously contrasts with basic models of informed lobbying in which lobbies' objective functions tend to be directly included in the social welfare. We, nonetheless, believe this simplification is an interesting (and technically tractable) starting point for understanding how information asymmetries affect the trade-off between centralization and decentralization. We assume that in each entity i the lobby's type θ_i is i.i.d. For the "aligned preferences" case, both θ s are drawn from a uniform distribution $f(\theta) = 1/(\bar{\theta} - \underline{\theta})$ on the interval $[\underline{\theta}, \bar{\theta}]$ with $3\underline{\theta} > \bar{\theta}$. For the "polarized preferences" case, the distribution of lobby A's type is identical to the aligned case, while lobby B's type is uniformly distributed within the interval $[-\bar{\theta}, -\underline{\theta}]$. We begin the analysis with the decentralized structure.

Decentralization.—In a decentralized structure, each lobby offers contributions to the policymaker of her entity. The political game is thus an informed principal problem. Since the policymaker does not care directly about the lobby's type, we have a private value informed principal problem. In such models, the lobby's private information does not affect the policymaker's decision. The policymaker only cares about whether the contribution compensates for shifting him away from his preferred policy. Hence, the policymaker does not take into direct account whether or not the contribution is revealing.⁴ Additionally, different lobbies' types prefer different policies, so it is not optimal to offer pooling contributions. This results in a political game where the lobby has no incentives to withdraw information. Hence, we can focus on informative contributions. As a consequence, there are no distortions due to information asymmetry, and the equilibrium policies are the same as in the perfect information decentralized structure, namely $\tilde{p}_i(\theta_i)$ given by (4).

Centralization.—In a centralized structure, lobbies offer contributions to the same policymaker. Remember that we follow the standard influence lobby group literature (Bernheim and Whinston 198, Grossman and Helpman 1995) that views the determination of policymaking as the outcome of a common agency game. Different lobbies (principals) use contributions as an incentive device to induce the

⁴In informed principal problems, the principal can benefit from delaying information revelation if the agent's decision depends on the realization of a random variable. This requires more complex contracts that allow the policy to be contingent on the information the principal reveals after the contract's acceptance. These complex contracts introduce an *ex ante* uncertainty to the agent at the moment of acceptance of the contract. Depending on the agent's utility, this uncertainty can help relax his individual rationality and incentive compatibility constraints (these constraints only need to hold in expected term, instead of state-by-state), increasing the surplus of their relationship. Such contracts do not reveal the lobby's private information directly, but the principal would reveal her private information before the implementation of the policy. Nonetheless, Maskin and Tirole (1990) show that revelation delaying creates no surplus gain when preferences are quasi-linear, which is the case we analyze here. That is, there is no benefit in offering a contract with revelation delay.

policymaker (agent) to make specific policy choices. The introduction of asymmetry of information between informed principals and uninformed agents, however, requires a specific strategy to find equilibrium policies.

Each lobby is privately informed about the realization of his type and does not know the rival's type. Therefore, the utility maximization problem of each lobby can be tackled as an informed principal problem with the policymaker. Several remarks are in order. First, in this informed principal problem, each lobby has private information about his own type while the policymaker has no direct private information. The policymaker, however, simultaneously receives the contributions from both lobbies. When contributions are separating, the policymaker learns the lobbies' types in equilibrium. Given that different types of lobby ask for different policy, the type of one lobby is relevant for the other lobby's payoff. Each lobby's problem becomes a principal-agent problem where the policymaker is privately informed about the rival's type.

Second, from Maskin and Tirole (1990), we know that informed principals do not gain from postponing information revelation. This justifies our focus on informative equilibria with separating differentiable contribution schedules.⁵ As a consequence, our political common agency game with exogenous asymmetric information between informed principals and an uninformed agent becomes, from the perspective of each principal, a principal-agent problem with an endogenously privately informed agent about the characteristics of the other principal.

We follow closely Martimort and Moreira (2010) and, as stated previously, we restrict ourselves to separating equilibrium strategies reflecting the fact that a given lobby i chooses different contribution schedules as his type changes.⁶ We first consider lobby i 's best-response contribution schedule to the rival lobby j 's strategy, assuming that the latter uses a separating strategy. Hence, before choosing the level of the joint policy p in the second stage of the game, the policymaker endogenously learns the private information by simply observing the contribution schedule proposed by the rival j . It follows that lobby i 's own optimal contribution schedule has to take into account the information rent that the policymaker obtains from his endogenous knowledge about θ_j . One may then characterize the optimal contribution schedule of lobby i , assuming that the policymaker is perfectly informed on lobby j 's type θ_j . As noticed by Martimort and Moreira (2010), the fact that the two lobbies' types do not enter directly into the policymaker's objective function ensures that the corresponding profile of contribution schedules is also a best response in the more general game where lobby i has asymmetric information on θ_i .⁷ Applying this approach, it turns out that lobby i 's best response is itself separating and therefore conveys information on his type to the policymaker. This observation then justifies

⁵This is also in the spirit of equilibrium allocations that are informative as in Spence (1973) and Riley (1979).

⁶In Section V and the online Appendix, we discuss the existence and implications of (non-informative) pooling equilibria of our lobbying game.

⁷The reason is that the incentive and participation constraints of the policymaker do not depend on his beliefs about lobby i 's type but only on the schedule that this lobby offers to him. Therefore, it follows that the policymaker's decisions to enter into the bilateral coalition with lobby i and to implement the policy p accordingly are also independent on his beliefs about the lobby's type. Any deviation from the contribution that lobby i would optimally offer if the policymaker were informed about his type is dominated for any off equilibrium path beliefs.

that the same techniques can be used to compute the rival lobby j 's best response. This approach consistently characterizes the informative equilibria we are seeking.

Specifically, we denote the realization of the type of district i lobby by θ_i and the realization of rival lobby j by θ_j . Solving backwards, given that we are in a separating equilibrium, the policymaker's problem has full knowledge about θ_i and θ_j when deciding his policy p . Given the separating contribution schedules $C_i(p, \theta_i)$ and $C_j(p, \theta_j)$, he then solves

$$(8) \quad \max_p C_i(p, \theta_i) + C_j(p, \theta_j) + \lambda W(p),$$

where we denote the utilitarian welfare of both entities by $W(p) = W_A(p) + W_B(p)$. This problem has the following first-order condition:

$$(9) \quad \frac{\partial C_i}{\partial p}(p, \theta_i) + \frac{\partial C_j}{\partial p}(p, \theta_j) + \lambda W'(p) = 0.$$

It is important to note that the equilibrium policy depends on the slopes $\partial C_i / \partial p$ and $\partial C_j / \partial p$ of the contribution schedules, which in turn, depend on the lobbies' types θ_i and θ_j . It follows that the equilibrium policy $p(\theta_i, \theta_j)$ satisfying (9) depends as well on the lobbies' types. Moreover, when the necessary second-order conditions of problem (8) hold and the contribution schedules $C_i(p, \theta_i)$ satisfy the Spence-Mirrlees property⁸ $\partial^2 C_i / \partial \theta_i \partial p \geq 0$ and $\partial^2 C_j / \partial \theta_j \partial p \geq 0$, simple differentiation of (9) provides that the equilibrium policy $p(\theta_i, \theta_j)$ is increasing with respect to the lobbies' types θ_i and θ_j in the "aligned preferences" setting. A similar argument shows that the equilibrium policy $p(\theta_A, \theta_B)$ is increasing with respect to θ_A and decreasing with respect to θ_B in the "polarized preferences" setting.

Consider each lobby's utility maximization problem. Since equilibrium policies are monotonic functions of the lobbies' types, the problem of choosing a contribution schedule and a policy can be reduced for each lobby i to the problem of choosing a value $\hat{\theta}_i$ that defines the slopes of the contributions, given (9) and the lobby's true type θ_i . Moreover, lobby i chooses her contribution non-cooperatively, uninformed about her rival's type θ_j . Therefore, she solves the following problem:

$$(10) \quad \max_{\hat{\theta}_i} E \left[-\frac{1}{2} [p(\hat{\theta}_i, \cdot) - \theta_i]^2 - C_i(p(\hat{\theta}_i, \cdot), \hat{\theta}_i) \right],$$

subject to (9), where the expectation is with respect to the rival type.

The focus on informative (truthful) strategies implies that the solution of (10) should be $\hat{\theta}_i = \theta_i$, for all $\theta_i \in [\underline{\theta}, \bar{\theta}]$. Following Martimort and Moreira (2010) and focusing on pointwise optimization, we obtain the following proposition characterizing the necessary optimality conditions of each type of lobby. All proofs are relegated to the Appendix.⁹

⁸We have to check that this condition holds after computing the equilibrium contributions $C_A(p, \theta)$ and $C_B(p, \theta)$.

⁹The explicit closed form solution of the equilibrium policies of Proposition 1 are computed in the online Appendix.

PROPOSITION 1: *The optimality conditions of (10) for lobby i are given by the first-order condition*

$$(11) \quad -[p - \theta_i] + \frac{\partial C_j}{\partial p}(p, \theta_j) - \lambda[2p - (\alpha_A + \alpha_B)] + H(\theta_j) \frac{\partial^2 C_j}{\partial \theta_j \partial p}(p, \theta_j) = 0,$$

for all $i, j \in \{A, B\}$, $i \neq j$, and $(\theta_i, \theta_j) \in [\underline{\theta}, \bar{\theta}]^2$, where $H(\theta_j)$ is the hazard rate¹⁰ and the second-order condition is

(i) for the “aligned preferences” case

$$\frac{\partial p}{\partial \theta_i}(\theta_i, \theta_j) \geq 0,$$

for all $i, j \in \{A, B\}$, $i \neq j$, and $(\theta_i, \theta_j) \in [\underline{\theta}, \bar{\theta}]^2$, and

(ii) for the “polarized preferences” case

$$\frac{\partial p}{\partial \theta_A}(\theta_A, \theta_B) \geq 0 \text{ and } \frac{\partial p}{\partial \theta_B}(\theta_A, \theta_B) \leq 0.$$

These first-order conditions given in Proposition 1 are standard in screening models. They state that the marginal surplus of the bilateral coalition between lobby i and the policymaker on the left side of (11) is equal to the marginal cost of the latter’s informational rent on the right side of (11). It looks similar to the first-order condition obtained under perfect information (5), except for the fact that there is now a new term due to the information distortion. Since lobby i does not know her rival’s type θ_j , she has to give incentives to the policymaker to report this type truthfully. This means that she has to screen the rival’s information from the policymaker. As in most screening problems, information rents must be given to ensure the policymaker reveals this piece of information and chooses a policy p according to the true type of the rival. To save on such rents enjoyed by the high-type rivals in the aligned setting, lobby i distorts the policy it demands when facing low-type rivals, reducing the slope of his contribution schedule with respect to the policy. The second-order condition requires only that the policy is increasing with respect to the lobby’s own type. This will be obtained when the second-order conditions of (8) are satisfied and the equilibrium contribution schedules $C_A(p, \theta)$ and $C_B(p, \theta)$ satisfy a Spence-Mirrlees property $\partial^2 C_A / \partial \theta_A \partial p \geq 0$ and $\partial^2 C_B / \partial \theta_B \partial p \geq 0$.

To compute the equilibrium policy p^* and the equilibrium informative contribution schedules $C_A(p, \theta)$ and $C_B(p, \theta)$, we solve the system of first-order conditions (11) together with the policymaker’s first-order condition (9). These equations define a system of partial differential equations in the contribution schedules $C_A(p, \theta)$ and $C_B(p, \theta)$ with boundary conditions given by the policymaker’s binding

¹⁰The hazard rate depends on the preferences being aligned or polarized. See the proof of Proposition 1 and the discussion below for more details.

participation constraint (no informational rent) for low types $\theta_i = \theta_j = \underline{\theta}$. For the aligned case, the hazard rate is $H(\theta_j) = (1 - F(\theta_j))/f(\theta_j) = \bar{\theta} - \theta_j$. The second-order conditions then can be checked ex post in the computed equilibrium. Martimort and Moreira (2010) show that a solution exists to this system for the symmetric case. We can compute an explicit expression for the equilibrium policy in the “aligned preferences” case

$$(12) \quad p^*(\theta_i, \theta_j) = \frac{\frac{3}{2}(\theta_i + \theta_j) - 2\bar{\theta} + \lambda(\alpha_A + \alpha_B)}{2(1 + \lambda)} = \bar{p}(\theta_i, \theta_j) - \frac{2\bar{\theta} - (\theta_i + \theta_j)}{4(1 + \lambda)},$$

where $\bar{p}(\cdot)$ is the centralized policy under perfect information, presented in (6). Notice that, in general, the equilibrium policy is smaller than $\bar{p}$, with equality holding only when both lobbies are of the high type (i.e., $\theta_i = \theta_j = \bar{\theta}$). Information asymmetry reduces the joint policy implemented by the policymaker. Moreover, the second-order condition as well as the Spence-Mirrlees condition hold. The full expressions for the contributions are in the online Appendix.

The intuition for these results follows from the fact that, at a best-response, each lobby induces a lower policy level than what would be ex post efficient for their bilateral coalition with the policymaker. This downward distortion reduces the information rent that the policymaker obtains from his endogenous private knowledge of the other lobby's type. Since both lobbies frame their contribution schedules in such a way, the actual equilibrium policy is reduced compared to the one obtained under perfect information. We refer to this effect as the *information transmission* effect. It endogenously creates informational advantage that the policymaker can exploit, therefore increasing the cost of influence of the lobbies. As the latter reduces the intensity of their contributions, policy shifts toward the efficient one. The *information transmission* effect brings a new perspective to the design of decision making under political influence. In a context of asymmetric information, centralization through delegation to a common policymaker creates a mechanism that provides informational leverage for the policymaker against interest groups. This results in less influence and reduced policy distortions.

The way that information asymmetry between lobbies and a policymaker is translated into the two principal agent problems where the agent is informed is very similar when preferences are polarized. The consequences of information transmission are, however, quite different. Remember that under perfect information, a lobby only exerts influence if she is stronger than the rival lobby. The distortions due to screening the rival's private information from the policymaker induces each lobby to exert less influence. The reduction is, nonetheless, less stringent when the lobbies face lower type rivals, while it is more stringent when they face relatively higher type rivals. Conversely, when a high type lobby faces a low type lobby, she is now facing a relatively weaker opponent. She is then able to exert more influence. Information transmission amplifies the relative differences in the lobbies' strengths. Therefore, the stronger lobby exerts more influence.

To compute the expression for $p^*(\theta_i, \theta_j)$, we follow the same steps as in the aligned case. The main difference is the hazard rate, now given by $H(\theta_j) = F(\theta_j)/f(\theta_j) = \theta_j - \underline{\theta}$. The explicit expression for the policy in this context is given by

$$(13) \quad p^*(\theta_i, \theta_j) = \frac{\frac{3}{2}(\theta_i - \theta_j) + \lambda(\alpha_A + \alpha_B)}{2(1 + \lambda)} = \bar{p}(\theta_i, \theta_j) + \frac{\theta_i - \theta_j}{4(1 + \lambda)},$$

where $\bar{p}(\cdot)$ is the centralized policy under perfect information for the polarized setting, presented in (7). It is straightforward to see that lobbies' characteristics difference $\theta_i - \theta_j$ has a greater weight in this expression than it does in the symmetric information policy (7). The algebra leading to this expression is presented in the online Appendix.

This common agency influence game with polarized preferences has a setting analogous to Martimort and Semenov (2008). They consider a privately informed agent about his ideology (the policymaker) and find that ideology uncertainty makes the equilibrium policy closer to the policymaker's bliss point. Here, we consider the polar case of privately informed principals (the lobbies), and we find opposite results. Inverting the source of asymmetric information also reverses the results: information asymmetry leads to greater political influence in our case.

III. Comparing Centralization and Decentralization

In this section, we compare centralized and decentralized structures in terms of welfare. Welfare is defined as the sum of the entities' welfare functions, $W(\cdot) = W_A(\cdot) + W_B(\cdot)$. This criterion excludes the payoffs of the players of the political game. This is equivalent to assume that the lobbies' and the policymaker's sizes are negligible compared to the society.

Given that we wish to highlight the role of lobbies' information asymmetries and the importance of the *information transmission* effect in the comparison between centralized and decentralized structures, we simplified the way the two entities A and B interact. Indeed, we only included the fact that centralized decision making tends to produce policies less responsive to the local environment than decentralized decision making (i.e., under a uniformity assumption the policies are exactly constrained to be the same). We should keep in mind, however, that this setting avoids other important dimensions discussed in the literature. In particular, our framework does not include features such as environmental externalities or strategic delegation across entities. Those elements are known to be important determinants of the comparison between centralized and decentralized structures.

Under perfect information, our model reproduces many effects that already have been emphasized under different forms in the literature. Namely, the aforementioned *uniformization* effect, *preference dilution* effect (in the aligned setting), and *competition* effect (in the polarized setting). As emphasized by the traditional literature on centralization (Oates 1972), the *uniformization* effect favors decentralization. The size of the welfare loss is directly related to the extent of difference between the entities' preferences, $\alpha_A - \alpha_B$.

The introduction of asymmetric information allows us to highlight a new effect into this trade-off: the *information transmission* effect. Interestingly, this effect reduces the vulnerability of policymaking to political influence in centralized systems in the “aligned preferences” context, but conversely, it increases political capture in the “polarized preferences” context. We assume, without loss of generality, $\Delta\alpha = \alpha_A - \alpha_B > 0$ and denote $\Delta\theta = \bar{\theta} - \underline{\theta}$. We start by discussing the “aligned preferences” context, comparing the perfect information benchmark case to the asymmetric information one. Then we repeat in a following subsection the same exercise for the “polarized preferences” case.

A. Aligned Preferences

Perfect Information.—In addition to the uniformization effect of local policies, centralization induces lobbies to offer contributions to the same (unique) policymaker, which creates another effect on the competition for influence. As a matter of fact, the policymaker cannot fully adjust his policy to reflect the preference of one specific lobby. He has to set the policy according to the “mix of political preferences” of the different interest groups he faces. We refer to this effect as a *preference dilution* effect. This effect was first illustrated in the political economy of centralization and regional agreements by de Melo, Panagariya, and Rodrik (1993).

The *preference dilution* effect increases social welfare by itself. Indeed, because the welfare function is concave in the policy level p , welfare associated with the average of the two distinct policy levels is greater than the average welfare of these policy levels. Hence, centralized policymaking that is subject to an “average political influence” of two lobbies generates higher social welfare for the two entities than decentralized policymaking subject to the political influence. Moreover, this effect increases with the range of lobby types, $\Delta\theta$, which determines the probability of having distinct lobbies across entities. In this perfect information setup, the trade-off between centralization and decentralization compares the two following effects: the *preference dilution* effect that favors centralization, and the *uniformization* effect that favors decentralization. Given that the *uniformization* effect (or conversely the *preference dilution* effect) is positively related to $\Delta\alpha$ (or conversely, to $\Delta\theta$), the trade-off depends on the relative sizes of $\Delta\theta$ and $\Delta\alpha$. Specifically, we can compute a threshold that relates these two key parameters.

PROPOSITION 2: *Suppose that types are perfectly informed and preferences are aligned. For every $\Delta\alpha$ there exists a threshold value $\widetilde{\Delta\theta}$ such that the expected social welfare under centralization is higher than under decentralization if and only if $\Delta\theta \geq \widetilde{\Delta\theta}$.*

Asymmetric Information.—Under centralization, Section II tells us that the lobbying game between the two interest groups and the joint policymaker generates an *information transmission* effect. This effect reduces the equilibrium level of the centralized policy. Given that lobbies intrinsically have preferences biased toward excessively large policy levels, this effect contributes positively to social welfare

under centralization. There is no such effect under decentralization. The *information transmission* effect is directly related to the degree of asymmetric information that exists between the lobbies and the policymaker. Therefore, it depends positively on the range of lobby types, $\Delta\theta$, and has no impact in this model when $\Delta\theta = 0$. Consequently, the *information transmission* effect provides an additional component of the trade-off between decentralization and centralization that favors the latter. For parameter configurations that make the two decision-making structures socially equivalent under perfect information, it shifts the trade-off in favor of centralization. The *information transmission* effect adds to the two preceding *uniformization* and *preference dilution* effects already identified under perfect information.

As under perfect information, the trade-off depends on the relative sizes of $\Delta\theta$ and $\Delta\alpha$. Clearly there is a configuration of these two parameters such that social welfare under centralized and decentralized systems are the same. Departing from this situation, a larger value of $\Delta\theta$ strengthens the *preference dilution* and the *information transmission* effects, and therefore makes centralization superior. On the other hand, a larger value of $\Delta\alpha$ reinforces the *uniformization* effect and therefore makes decentralization superior. Moreover, when lobbies have private information, that same configuration of parameters is more likely to induce centralization than with perfect information. This sums up to the following result.

PROPOSITION 3: *Suppose that types are private information. For every $\Delta\alpha$, there exists a threshold value $\widehat{\Delta\theta}$ such that the expected social welfare under centralization is higher than under decentralization if and only if $\Delta\theta \geq \widehat{\Delta\theta}$. Moreover, for every $\Delta\alpha$, we have that $\widehat{\Delta\theta} > \widehat{\Delta\theta}$.*

Proposition 3 is quite similar to Proposition 2. The thresholds, however, are not the same. Comparing the two propositions shows how *information transmission* affects the trade-off between centralization and decentralization. The last statement of Proposition 3 shows that the range of parameters in which centralization is welfare superior is larger with private information. That is, we can find combinations of $\Delta\theta$ and $\Delta\alpha$ such that centralization is not welfare superior under perfect information while it is welfare superior with asymmetric information (while the opposite, however, cannot occur). The bottom line of this comparison is the fact that centralization is more likely to be welfare superior with private information in the “aligned preferences” context.

B. Polarized Preferences

With polarized preferences the interaction between lobbies in centralized structures is quite different from the aligned context. The policymaker’s preferences are now in the middle ground between the two lobbies’ preferences. The influence of one lobby pushes policies further away from the rival’s bliss point. This creates strong competitive pressures between lobbies on the market for influence.

Perfect Information.—While in the aligned setting each lobby benefits from the influence of the other, now the influence of a lobby harms the rival. Each lobby

desires to neutralize the rival's influence and, on top of that, to influence the policy-maker. As a consequence, the policy is biased toward the preference of the stronger lobby. This is the *competition* effect. With lobbies of equal strength, no effective influence is exerted and the policy follows the districts' preferences.

In this setting, the trade-off between centralization and decentralization weights this *competition* effect against the *uniformization* effect. The larger $\Delta\theta$, the larger the scope for disagreement is among the lobbies; the stronger the *competition* effect, the better it is to have a centralized structure. However, much like in the "aligned preferences" case, the larger $\Delta\alpha$, the stronger is the *uniformization* effect, and the worse it is to implement centralization. Interestingly, the value of $\Sigma\theta := \bar{\theta} + \underline{\theta}$ also matters for welfare comparison. It reflects a measure of the relative distance between the average ideal policy level that each lobby would like to see implemented.¹¹ When the lobbies' average bliss point is much higher than the society's, they exert a lot of influence in decentralized structures. Conversely, in centralized structures, they are forced to compete strongly as their positions are on average highly opposed. As a consequence, their influence gets mitigated. The greater the lobbies' average ideal policy strength is compared to the α s, the more likely it is that centralization is beneficial. In other words, the larger $\Sigma\theta$, the more likely centralization is welfare superior. These observations are summarized in the following proposition.

PROPOSITION 4: *Suppose that types are perfect information. For every $\Delta\alpha$:*

- (i) *there exists a threshold value $\widetilde{\Delta\theta}$ such that the expected welfare under centralization is higher than under decentralization if and only if $\Delta\theta \geq \widetilde{\Delta\theta}$;*
- (ii) *there exists a threshold value $\widetilde{\Sigma\theta}$ such that the expected welfare under centralization is higher than under decentralization if and only if $\Sigma\theta \geq \widetilde{\Sigma\theta}$.*

Asymmetric Information.—The information transmission effect has quite different implications with polarized preferences. When designing its contribution, each lobby would ideally like to face a moderate opponent, where there is less opposition to her own influence. Importantly, though, the policymaker has a natural tendency to lie to the lobby by overstating the policy position of his opponent. The policymaker does so in order to receive the greater marginal contribution that the lobby would be ready to offer to neutralize the opponent's position. Therefore, to satisfy the incentive compatibility constraint of the policymaker, the lobby has to increase the contribution he pays when facing a low-type opponent. To save on this higher contribution, he reduces the marginal policy he requests when facing a high-type opponent. As a consequence, strong opponents face weaker opposition, which induces more influence when the lobbies' average ideal positions are unbalanced.

¹¹ Lobby A's type is uniformly distributed between $[\underline{\theta}, \bar{\theta}]$, while lobby B's type is uniformly distributed between $[-\bar{\theta}, -\underline{\theta}]$. Lobby A's average ideal policy level is simply $\int_{\underline{\theta}}^{\bar{\theta}} \theta \cdot \frac{d\theta}{\Delta\theta} = \frac{\Sigma\theta}{2}$, while lobby B's ideal average policy level is $\int_{-\bar{\theta}}^{-\underline{\theta}} \theta \cdot \frac{d\theta}{\Delta\theta} = -\frac{\Sigma\theta}{2}$. So the difference between the two lobbies' average ideal policy levels is exactly $\Sigma\theta$.

In other words, information transmission weakens the lobbies' competition, making it less effective. Notice, however, that competition will still be effective when the lobbies' types are balanced. This indicates that for a given asymmetry between districts preferences $\Delta\alpha$, there still exist ranges of lobbies' types that make centralization welfare superior. Specifically, we have the following.

PROPOSITION 5: *Suppose that lobbies' types are private information. For every $\Delta\alpha$:*

- (i) *there exists a threshold value $\widehat{\Delta\theta}$ such that the expected welfare under centralization is higher than under decentralization if and only if $\Delta\theta \geq \widehat{\Delta\theta}$;*
- (ii) *there exists a threshold value $\widehat{\Sigma\theta}$ such that the expected welfare under centralization is higher than under decentralization if and only if $\Sigma\theta \geq \widehat{\Sigma\theta}$;*
- (iii) *we have that $\widehat{\Delta\theta} > \widetilde{\Delta\theta}$ and $\widehat{\Sigma\theta} > \widetilde{\Sigma\theta}$.*

Statements (i) and (ii) from Propositions 4 and 5 are quite similar. The difference is stressed in statement (iii) of Proposition 5. It shows that centralization is welfare superior for a smaller range of parameters when lobbies have private information. Indeed, under centralization, the *information transmission* effect has a negative impact on welfare. It reduces the effectiveness of the *competition* effect that naturally arises in perfect information when lobbies have polarized preferences. This leads to greater capture by the more extreme lobby.

IV. Examples

In this section, we briefly illustrate our previous results in two simple examples: the classical problem of local public goods provision across communities or districts and the formation of custom unions in international trade. We provide in the main text an informal intuitive discussion of the arguments, while the formal details appear in the online Appendix.

A. Local Public Goods Provision

Much of the political economy literature that goes beyond Oates' classical trade-off between centralization and decentralization only considers situations with symmetric information between interest groups and policymakers. Our framework allows us to reconsider this classical problem in the presence of lobbies who are privately informed about their abilities to exert influence. This setup also allows us to discuss how different structures of public budgets affect the lobbying competition and, ultimately, the provision of local public goods.

Two districts have to decide how much of a local public good to provide. The lobbies represent organized members of the society with higher valuation for the public good, such as, for instance, organized elites with preferences not aligned with the

average citizen.¹² We consider three alternative institutional structures. In the first one, the decision is decentralized and public budgets across districts are separate. In the second structure, the district budgets are still separate, but the policy decision is centralized and constrained to be uniform. Finally, we study a centralized decision structure without policy uniformity, while there is a two-stage budgeting process that is integrated across districts (a common pool financing).¹³

When the public good decision is decentralized and budgets are separate, our previous results from the decentralized structure apply. The model is solved as a principal-agent game in which the policymaker only takes into consideration the welfare of his own district. Information asymmetry has no stake and the solution is efficient for the relationship between a lobby and the policymaker. The public good is provided up to the point where the lobby's marginal benefit equals the marginal cost of provision.

Under centralization with uniform public good provision and separate district budgets, lobbies have "aligned preferences" as both want more of their own public goods and offer contributions to the same policymaker.

Under a centralized system with a nonuniform public good decision and a two-stage integrated budget process with a fixed common pool budget, lobbies have polarized preferences. As the budget is now already set when lobbies offer contributions to the policymaker, what a lobby can get in terms of his district-specific public good is just what the other lobby cannot get for his own district.

How do the three institutional structures described above compare in terms of incentives and welfare? In a decentralized structure, each district's specific lobby exerts uncontested influence, but the policy choice reflects better the district's social preferences. With uniform centralized public good provision, the policymaker takes into account the average preference of the two districts. In such a setting with independent budgets, lobbies have aligned interests. Private information makes their coordination more difficult by giving extra bargaining power to the policymaker. This, in turn, favors the use of a uniform policy in terms of welfare.

A common pool two-stage integrated budget calls also for a centralized decision, possibly with some degree of flexibility with respect to local policies. In such structure, lobbies have polarized preferences and compete strongly for the public good allocation. Their political influence is weakened because of this competition. Private information, however, hinders this *competition* effect and to stimulate political capture by the more extreme lobby. This suggests that budget centralization with flexible policies has some clear political economy advantage compared to uniform centralized decisions and independent budgets under perfect information. However, the latter institutional setting may recover some advantages as the information problem on the lobbies' side becomes more stringent.

¹² Our results would be similar if lobbies had lower valuation for the public good.

¹³ An alternative structure would combine a centralized decision without uniform policy and with a decentralized budget. In the absence of interdistrict externalities, this last structure is identical to a decentralized one.

B. *Tariff Protection in Customs Unions*

Our model of political influence under asymmetric information also allows us to reconsider the issue of preferential trade agreements (PTAs) when there is private information on the side of protectionist lobbies. Specifically, consider two small open countries A and B that trade with the rest of the world. In each country, there is a protectionist lobby that employs resources to obtain tariff protection against foreign imports. Without a trade agreement between the two countries, each lobby demands protection from the government of her own country. In a customs union, the lobbies from both countries compete to influence the common level of trade protection decided by the customs union. Additionally, lobbies have private information about their political strength as a group.

More precisely, consider a simple partial equilibrium model with a good x that can be imported by both countries A and B .¹⁴ Demand for that good is linear in each country. Additionally, the good is produced in each country using labor and a specific factor that is in limited supply. Consequently, producers have capacity constraints.

Each government collects import taxes and the sum of the firm's profits, consumer surplus, and the government's tariff revenue gives the welfare of the society. A political influence game takes place within each economy. The lobby of each country offers contributions to the policymaker in order to influence the tariff decision. Each economy has a lobby that represents the producers of good x . Lobbies are "principals" of the political game and policymakers are agents in the political game.

The two countries may form a customs union or retain a non-coordinated trade policy. Without the agreement, the policy decision is decentralized. Each country delegates its trade policy decision to a national policymaker who chooses the import tariff of the economy or, equivalently, the economy's domestic price. If the two countries sign a customs union agreement, they delegate the policy choice to a single policymaker who is restricted to setting a uniform policy (the tariffs of the two economies are the same). This is a centralized decision-making setting.

In this setting, it can be shown that each country's welfare function takes a simple quadratic form and that the lobbies' preferences are linear in the tariff policy and therefore characterized by an infinite bliss point. Although the setup is slightly different from our full quadratic framework presented in Section I, this is yet a case of aligned preferences, and the results are similar to our baseline model. It can be shown, in particular, that there is less protection in a customs union agreement when lobbies have private information. The intuition is that the *information transmission* effect increases the cost of political influence, granting more bargaining power to the policymaker. As a consequence, lobbies' private information under a customs union leads to a fall of protection and an increase in imports and social welfare.

¹⁴ There is also a numeraire good produced from labor only in a one-to-one rate of transformation.

V. Extensions

An important feature of our lobbying game is to follow a standard political economy approach (PEA) in which lobbies act as principals and the policymaker is an agent subject to their influence for the choice of his policy variable. From a normative point of view, it may be therefore natural to ask how this perspective compares to a mechanism design approach (MDA) in which we keep the same basic structure but reverse the bargaining power between the policymaker and the lobbies. This is the purpose of the next subsection.

In addition, to highlight the existence and the role of screening and the information transmission effect in the trade-off between centralization and decentralization, our analysis so far focuses on separating (informative) equilibria of the lobbying game under centralization. As is well known in multiple privately informed principals games, other non-informative equilibria may also exist. Therefore, the following section investigates the existence of a pooling equilibrium and, when it exists, we discuss its implications compared to the separating equilibrium we focused on in this paper.

A. Mechanism Design Approach

The policymaker is now the principal, and the lobbies are the informed agents. The game setting is modified accordingly. Under decentralization, each district has its own policymaker who proposes a contract to the privately informed lobby. Under centralization, a common policymaker proposes a contract to two privately informed lobbies.

Under the decentralized structure, the lobbying game becomes a standard relationship between a uniformed principal and a privately informed agent. The policymaker trades informational rents for more favorable policies. Differently from the common agency game, the lobby refrains to reveal her private information directly to the policymaker. As in standard principal-agent problems, information distortions arise and, except at the top, equilibrium policies are smaller than the ones obtained when lobbies propose the contract. The MDA leads to more information distortions but enhances the social welfare since policies are closer to the district's preferred policy.

Under centralized decision making, the common policymaker proposes a contract to two informed lobbies restricted to uniform policy. With aligned preferences, information distortions are more pronounced under the MDA. The nature of the screening problem in this setting explains this result. Under the PEA, each lobby screens the policymaker's marginal cost of providing a favorable policy. This marginal cost indirectly depends on the rival lobby's type, so each lobby is screening the rival's information. With the MDA, the policymaker is now directly interested in lobbies' types, which determine their willingness to pay for a favorable policy. As a consequence, information screening leads to greater information distortions than under the PEA.

In this rent-seeking setting with aligned preferences, information distortions are positively associated with welfare. So, reversing the bargaining power between

policymakers and lobbies is social welfare enhancing. In addition, information distortions under centralization are an average of the information distortions of the decentralized setting. The difference between centralized and decentralized structures is only due to the uniformization effect, which averages out the society's preferences, the lobbies' influence, and the information distortions.

With polarized preferences, information distortions are once again more pronounced in the MDA when compared to the PEA. Information distortions, however, tend to undermine lobbies' competition, resulting in a policy even further away from the welfare optimum. As a consequence, centralization leads to a lower social welfare level under the MDA when compared to the PEA.

From a normative point of view, the comparison of the benefits of centralization relative to decentralization according to the PEA and to the MDA depends on the lobbies' preferences. When the preferences are aligned, aggregate welfare for both (decentralized and centralized) regimes is larger under the MDA than under the PEA. However, when preferences are polarized, the MDA leads to larger aggregate welfare gains for the decentralization regime and smaller welfare gains for the centralization regime. As a consequence, centralization is less likely to dominate decentralization under the MDA rather than under the PEA.

B. *Pooling Equilibria*

As previously stated, our lobbying game with multiple privately informed principals may also possess other non-informative equilibria. In this subsection, we verify whether we can construct pooling equilibria and how they compare with the separating equilibrium analyzed in this paper.

In such equilibria, all types of a given lobby pool and offer the same contract that specifies a contribution for a given policy target. The policymaker cannot learn anything from observing the contribution, and the other lobby has nothing to screen about and is then forced to agree on this policy target.

In the online Appendix, we provide conditions for the existence of a pooling equilibrium. In general, the pooling equilibrium may lead to more or less distortions than the separating equilibrium. If lobbies' preferences are aligned, on the one hand, lobbies do not have to leave informational rents to the policymaker to screen the information the latter obtains in a separating equilibrium. This effect reduces the cost of lobbying and, therefore, increases the lobbying influence and reduces the welfare by driving the equilibrium policy away from the socially optimal policy. On the other hand, in a pooling equilibrium, lobbies have to implement a policy that is independent from their own type. Consequently, their influence over the policymaker is reduced, which is welfare improving. From these two effects, it follows that, in general, there is no clear welfare ranking between the pooling and the separating equilibria. The online Appendix provides conditions under which the pooling equilibrium actually leads to lower distortions than the separating equilibrium and is welfare improving.¹⁵

¹⁵ Specifically if the common policymaker's bliss point is greater than the sum of the lobbies' highest possible type bliss points, then we can find a pooling equilibrium that leads to a lower distortion than the separating equilibrium.

For polarized preferences, analogous results can also be derived. The comparison between the pooling and the separating equilibria follows the same logic as in the aligned preferences case, with again the main difference that, because of the competition effect, pooling can give an advantage to the lobbies compared to the separating equilibrium and, therefore, more likely to be welfare reducing.

VI. Conclusion

In this paper, we considered the trade-off between centralized and decentralized policymaking when policymakers are subject to capture by privately informed lobbies. We identified a new *information transmission* effect in the political game under centralized structures that changes the way lobbies interact. When lobbies interests are aligned, information transmission reduces capture in centralized decision-making structures. The basic insight comes from two features. First, in centralized systems, policies tend to integrate cross-entity specificities and therefore create strategic informational interdependencies for privately informed lobbies. Each lobby's optimal influence strategy depends on privately known characteristics of rival lobbies. Second, centralization forces competition for political influence to be focused on one central policymaker. Since this competition is information revealing, the common policymaker has a privileged position to obtain valuable private information about each lobby's characteristics. Therefore, centralization allows the policymaker to enjoy information rents, increasing his bargaining capacity in bilateral relationships with interest groups. As a consequence, the cost of political influence increases and the extent of political capture declines.

When the lobbies' preferences are polarized, the opposite effect emerges. Centralized decision with polarized preferences tends to enhance lobbies' competition for influence, which in turn reduces their rent-seeking lobbying. The *information transmission* effect, however, mitigates this competition, therefore diminishing the benefits of centralization.

The framework we used to highlight this insight clearly abstracts from many dimensions relevant to the comparison between centralized and decentralized systems. Enriching the model would, however, generate greater strategic informational interdependence across lobbies than is necessary to get the *information transmission* effect. This reinforces the case for centralized systems.

More substantially, one may think about enriching the political process of policy determination under centralization. For example, one may think of more complex decision processes involving bargaining between district representatives, each subject to political influence. It would be interesting to see how such variations in the political structure interact with privately informed rent-seeking groups. Another worthwhile extension would be to allow the possibility of lobbies holding efficiency-improving private information. In that case, lobbying activity could play a positive social role that might affect the relative benefits of centralization.

Finally, beyond our simple examples on local public good provision and customs union agreements, other political economy contexts with the interplay between political influence competition and asymmetry of information may also generate rich and interesting insights for the optimal allocation of decision rights in public policy areas.

APPENDIX

PROOF OF PROPOSITION 1:

This proof presents the computations for obtaining the first-order condition of problem (10) in detail. The proof follows Martimort and Moreira (2010) closely. We begin with the “aligned preferences” setting.

Following the tradition of the literature, we eliminate the contribution from the objective function (10) before computing the best responses. We write

$$E\left[C(p(\hat{\theta}_i, \cdot), \hat{\theta}_i)\right] = \int_{\underline{\theta}}^{\bar{\theta}} C(p(\hat{\theta}_i, \theta_j), \hat{\theta}_i) f(\theta_j) d\theta_j,$$

and integrate it by parts to get

$$\begin{aligned} \int_{\underline{\theta}}^{\bar{\theta}} C(p(\hat{\theta}_i, \theta_j), \hat{\theta}_i) f(\theta_j) d\theta_j &= -(1 - F(\theta_j)) C(p(\hat{\theta}_i, \theta_j), \hat{\theta}_i) \Big|_{\underline{\theta}}^{\bar{\theta}} \\ &\quad + \int_{\underline{\theta}}^{\bar{\theta}} (1 - F(\theta_j)) \frac{\partial C}{\partial p}(p(\hat{\theta}_i, \theta_j), \hat{\theta}_i) \frac{\partial p}{\partial \theta_j}(\hat{\theta}_i, \theta_j) d\theta_j. \end{aligned}$$

In what follows, we omit the arguments of $p(\hat{\theta}_i, \cdot)$ and its derivatives for the sake of exposition. Substituting the policymaker first-order condition (9) in the equation above gives

$$E\left[C(p, \hat{\theta}_i)\right] = C(p(\hat{\theta}_i, \underline{\theta}), \hat{\theta}_i) - E\left[\left(\frac{\partial C}{\partial p}(p, \cdot) + \lambda W'(p)\right) \frac{1 - F(\cdot)}{f(\cdot)} \frac{\partial p}{\partial \theta_j}\right].$$

Then, inserting this last expression back into (10) leads to the following problem:

$$\max_{\hat{\theta}_i} E\left[-\frac{1}{2}(p - \theta_i)^2 + \left(\frac{\partial C}{\partial p}(p, \cdot) + \lambda W'(p)\right) \frac{1 - F(\cdot)}{f(\cdot)} \frac{\partial p}{\partial \theta_j}\right] - C(p(\hat{\theta}_i, \underline{\theta}), \hat{\theta}_i).$$

We now differentiate this expression to obtain the first-order condition of problem (10):

$$\begin{aligned} \text{(A.1)} \quad E\left[-(p - \theta_i) \frac{\partial p}{\partial \theta_i} + \left(\frac{\partial^2 C}{\partial^2 p}(p, \cdot) + \lambda W''(p)\right) \frac{1 - F(\cdot)}{f(\cdot)} \frac{\partial p}{\partial \theta_i} \frac{\partial p}{\partial \theta_j}\right] \\ + E\left[\left(\frac{\partial C}{\partial p}(p, \cdot) + \lambda W'(p)\right) \frac{1 - F(\cdot)}{f(\cdot)} \frac{\partial^2 p}{\partial \theta_i \partial \theta_j}\right] \\ - \frac{\partial C}{\partial p}(p(\hat{\theta}_i, \underline{\theta}), \hat{\theta}_i) \frac{\partial p}{\partial \theta_i}(\hat{\theta}_i, \underline{\theta}) - \frac{\partial C}{\partial \theta_i}(p(\hat{\theta}_i, \underline{\theta}), \hat{\theta}_i) = 0. \end{aligned}$$

We then integrate by parts the term in the second line of (A.1), which gives

$$\begin{aligned}
 (A.2) \quad & - \left(\frac{\partial C}{\partial p}(p, \hat{\theta}_i) + \lambda W'(p) \right) \frac{\partial p}{\partial \theta_i} \\
 & - E \left[\left(\left(\frac{\partial^2 C}{\partial^2 p}(p, \cdot) + \lambda W''(p) \right) \frac{\partial p}{\partial \theta_j} + \frac{\partial^2 C}{\partial \theta_j \partial p}(p, \cdot) \right) \frac{(1 - F(\cdot))}{f(\cdot)} \frac{\partial p}{\partial \theta_i} \right] \\
 & + E \left[\left(\frac{\partial C}{\partial p}(p, \cdot) + \lambda W'(p) \right) \frac{\partial p}{\partial \theta_i} \right].
 \end{aligned}$$

Substituting (A.2) back into the first-order condition (A.1) and using (9) gives

$$\begin{aligned}
 (A.3) \quad & E \left[\left(-(p - \theta_i) + \frac{\partial C}{\partial p}(p, \cdot) + \lambda W'(p) - \frac{\partial^2 C}{\partial \theta_j \partial p}(p, \cdot) \frac{1 - F(\cdot)}{f(\cdot)} \right) \frac{\partial p}{\partial \theta_i} \right] \\
 & - \frac{\partial C}{\partial \theta_i}(p(\hat{\theta}_i, \underline{\theta}), \hat{\theta}_i) = 0.
 \end{aligned}$$

Using a boundary condition for the contribution, we can further simplify this expression. The boundary condition comes from the policymaker's participation constraint. If the rival lobby is low type, the policymaker cannot lie to the lobby. The lobby, then, has no reason to leave rents. In such a case, the policymaker gets his reserve utility. That implies that

$$C(p(\hat{\theta}_i, \underline{\theta}), \hat{\theta}_i) + C(p(\hat{\theta}_i, \underline{\theta}), \underline{\theta}) + \lambda W(p(\hat{\theta}_i, \underline{\theta})) = \lambda W(p^e), \forall \hat{\theta}_i.$$

Differentiating this expression with respect to $\hat{\theta}_i$ gives

$$\left[\frac{\partial C}{\partial p}(p(\hat{\theta}_i, \underline{\theta}), \hat{\theta}_i) + \frac{\partial C}{\partial p}(p(\hat{\theta}_i, \underline{\theta}), \underline{\theta}) + \lambda W'(p(\hat{\theta}_i, \underline{\theta})) \right] \frac{\partial p}{\partial \theta_i}(\hat{\theta}_i, \underline{\theta}) + \frac{\partial C}{\partial \theta_i}(p(\hat{\theta}_i, \underline{\theta}), \hat{\theta}_i) = 0.$$

Using (9), it is straightforward to see that the last term of (A.3) is zero. This further simplifies the first-order condition (A.3) to

$$E \left[\left(-(p - \theta_i) + \frac{\partial C}{\partial p}(p, \cdot) + \lambda W'(p) - \frac{1 - F(\cdot)}{f(\cdot)} \frac{\partial^2 C}{\partial \theta_j \partial p}(p, \cdot) \right) \frac{\partial p}{\partial \theta_i} \right] = 0.$$

Additionally, given the concavity of the functional (10), the second-order condition of the problem is simplified to

$$E \left[\frac{\partial p}{\partial \theta_i}(\theta_i, \cdot) \right] = 0.$$

Following Martimort and Moreira (2010), we require truth-telling and focus on pointwise optimization. The first-order and second-order conditions become

$$(A.4) \quad -(p - \theta_i) + \frac{\partial C}{\partial p}(p, \theta_j) + \lambda W'(p) - \frac{1 - F(\theta_j)}{f(\theta_j)} \frac{\partial^2 C}{\partial \theta_j \partial p}(p, \theta_j) = 0,$$

$$\frac{\partial p}{\partial \theta_i} \geq 0,$$

for all $(\theta_i, \theta_j) \in \Theta^2$.

The computation for the polarized case is similar but not identical. The nature of lobbying competition is, however, different, which leads to different steps to eliminate the contribution from (10). The intuition is that, since lobbies have opposing preferences, they now desire to face low-type rivals. So the “bad” state of nature, where they leave zero rents for the policymaker, is $\bar{\theta}$. The departing point, nevertheless, is the same:

$$E[C(p, \hat{\theta}_i)] = \int_{\underline{\theta}}^{\bar{\theta}} C(p, \hat{\theta}_i) f(\theta_j) d\theta_j.$$

Then, we use integration by parts in a slightly different way to obtain

$$\int_{\underline{\theta}}^{\bar{\theta}} C(p, \hat{\theta}_i) f(\theta_j) d\theta_j = F(\theta_j) C(p, \hat{\theta}_i) \Big|_{\underline{\theta}}^{\bar{\theta}} - \int_{\underline{\theta}}^{\bar{\theta}} (F(\theta_j)) \frac{\partial C}{\partial p}(p, \hat{\theta}_i) \frac{\partial p}{\partial \theta_j} d\theta_j.$$

The following steps basically replicate the ones for the aligned setting. For the sake of exposition, we omit these steps. Again, we require truth-telling and focus on pointwise optimization. The first-order condition is given by

$$(A.5) \quad \begin{aligned} -(p - \theta_A) + \frac{\partial C}{\partial p}(p, \theta_B) + \lambda W'(p) + \frac{F(\theta_B)}{f(\theta_B)} \frac{\partial^2 C}{\partial \theta_B \partial p} &= 0, \\ -(p + \theta_B) + \frac{\partial C}{\partial p}(p, \theta_A) + \lambda W'(p) + \frac{F(\theta_A)}{f(\theta_A)} \frac{\partial^2 C}{\partial \theta_A \partial p} &= 0 \end{aligned}$$

for all $(\theta_A, \theta_B) \in \Theta^2$.

Notice that the only difference between (A.5) and (A.4) is the hazard rate. By denoting the hazard rates by $H(\theta_j)$, both (A.5) and (A.4) can be expressed as (11). This completes the proof. The online Appendix gives the continuation of this proof, which presents the derivation leading to the expression for equilibrium policies under centralization. ■

PROOF OF PROPOSITION 2:

In this proof, we compare the welfare of decentralized and centralized structures. We begin with the aligned setting under perfect information and we compute the expected value of the difference between the welfare of the two structures.

Remember each district's welfare is given by

$$W_i(p_i) = -\frac{1}{2}(p_i - \alpha_i)^2.$$

For centralized structures, we evaluate W_i at the policy under centralization (6). The expressions for welfare are

$$W_i(\bar{p}) = -\frac{1}{2} \left(\frac{\theta_A + \theta_B - 2\alpha_A - \lambda(\alpha_i - \alpha_{-i})}{2(1 + \lambda)} \right)^2,$$

where $i \in \{A, B\}$ and $i \neq -i$. We then sum the two expressions to compute the welfare of the entire economy:

$$(A.6) \quad W(\bar{p}) = -\frac{1}{4(1 + \lambda)^2} \left[(\theta_i + \theta_j)^2 - 2(\alpha_A + \alpha_B)(\theta_i + \theta_j) + \lambda(2 + \lambda) \Delta\alpha + 2(\alpha_A^2 + \alpha_B^2) \right].$$

We now compute the welfare in decentralized structures. Evaluating W_i at (4) gives

$$W_i(\check{p}_i) = -\frac{1}{2} \left(\frac{\theta_i + \lambda\alpha_i}{1 + \lambda} \right)^2,$$

where $i \in \{A, B\}$. Summing the two expressions, we compute the society's welfare,

$$(A.7) \quad W(\check{p}_A, \check{p}_B) = -\frac{1}{2(1 + \lambda)^2} \left[\theta_i^2 + \theta_j^2 - 2(\alpha_A\theta_i + \alpha_B\theta_j) + \alpha_A^2 + \alpha_B^2 \right].$$

We now compute the difference between (A.6) and (A.7):

$$W(\check{p}_A, \check{p}_B) - W_A(\bar{p}) - W_B(\bar{p}) = -\frac{1}{4(1 + \lambda)^2} \left[(\theta_i - \theta_j)^2 - 2(\theta_i - \theta_j) \Delta\alpha - \lambda(2 + \lambda) \Delta\alpha^2 \right].$$

The expected value of this expression is given by

$$\begin{aligned} E[W(\check{p}_A, \check{p}_B) - W(\bar{p})] \\ = -\frac{1}{4(1 + \lambda)^2 \Delta\theta^2} \int_{\underline{\theta}}^{\bar{\theta}} \int_{\underline{\theta}}^{\bar{\theta}} \left[\theta_i^2 - 2\theta_i\theta_j + \theta_j^2 \right. \\ \left. - 2(\theta_i - \theta_j) \Delta\alpha - \lambda(2 + \lambda) \Delta\alpha^2 \right] d\theta_i d\theta_j. \end{aligned}$$

Doing straightforward algebra, we get the following expression for this integral:

$$(A.8) \quad \Omega(\Delta\theta) = E[W(\check{p}_A, \check{p}_B) - W(\bar{p})] = -\frac{1}{4(1 + \lambda)^2} \left[\frac{\Delta\theta^2}{6} - \lambda(2 + \lambda) \Delta\alpha^2 \right].$$

Notice that the sign of Ω depends on the expression inside brackets. If Ω is positive, then decentralization is welfare superior. On the other hand, if Ω is negative,

centralization is welfare superior. Notice that Ω is decreasing with respect to $\Delta\theta$ and increasing with respect to $\Delta\alpha$. Moreover, for any given $\Delta\alpha$, take $\widehat{\Delta\theta} = \Delta\alpha\sqrt{6\lambda(1+\lambda)}$. Then, $\Omega(\widehat{\Delta\theta}) = 0$. For any $\Delta\theta > \widehat{\Delta\theta}$, Ω is negative and centralization is welfare superior, which proves the proposition. ■

PROOF OF PROPOSITION 3:

The proof of Proposition 3 follows closely the proof of Proposition 2. We compute the expected welfare difference between the decentralized and centralized structures under the aligned setting with asymmetric information.

The society's welfare in decentralized structures is the same as under perfect information (the policies are the same). The society's welfare under this structure is, thus, given by (A.7). For the centralized structure, the information transmission now affects the lobbies' strategies and, consequently, the equilibrium policy. Each district's welfare is given by

$$W_i(p^*) = -\frac{1}{2} \left(\frac{3(\theta_A + \theta_B) - 2\theta - 4\alpha_i - 2\lambda(\alpha_i - \alpha_{-i})}{4(1+\lambda)} \right)^2,$$

where $i \in \{A, B\}$ and $i \neq j$. The expected difference between the welfare of decentralized and centralized structures is given by the following expression:

$$\begin{aligned} & W_A(\check{p}_A) + W_B(\check{p}_B) - W_A(p^*) - W_B(p^*) \\ &= -\frac{1}{4(1+\lambda)^2 \Delta\theta^2} \\ &\quad \times \int_{\underline{\theta}}^{\bar{\theta}} \int_{\underline{\theta}}^{\bar{\theta}} \left[\theta_i^2 - 2\theta_i\theta_j + \theta_j^2 - \lambda(2+\lambda) \Delta\alpha^2 - 2(\theta_i - \theta_j) \Delta\alpha \right. \\ &\quad \left. - \left(\bar{\theta} - \frac{\theta_i + \theta_j}{2} \right)^2 + 2(\theta_i + \theta_j - (\alpha_A + \alpha_B)) \left(\bar{\theta} - \frac{\theta_i + \theta_j}{2} \right) \right] d\theta_i d\theta_j. \end{aligned}$$

Straightforward algebra leads to the following expression for this difference, which we denote by

$$(A.9) \quad \Psi(\Delta\theta) = -\frac{1}{4(1+\lambda)^2} \left[\frac{17\Delta\theta^2}{24} - \lambda(2+\lambda) \Delta\alpha^2 + \Delta\theta(2\theta - (\alpha_A + \alpha_B)) \right].$$

Again, a positive Ψ indicates decentralization is welfare superior while a negative Ψ indicates centralization is welfare superior. It is also straightforward to show that for any $\Delta\alpha$, there exists a $\widehat{\Delta\theta}$ such that $\Psi(\widehat{\Delta\theta}) = 0$. For any $\Delta\theta > \widehat{\Delta\theta}$, Ψ is negative and centralization is welfare superior. Finally, we may write

$$\Psi(\Delta\theta) = \Omega - \frac{1}{4(1+\lambda)^2} \left[\frac{13}{24} \Delta\theta^2 + \Delta\theta(2\theta - (\alpha_A + \alpha_B)) \right],$$

from which it is clear that $\Psi < \Omega$. The value $\widetilde{\Delta\theta}$ is, by definition, such that $\Omega(\widetilde{\Delta\theta}) = 0$. Since $\Psi < \Omega$ and Ψ is decreasing, then $\Psi(\widetilde{\Delta\theta}) < 0$ and centralization is welfare superior for $\widetilde{\Delta\theta}$. This implies that $\widetilde{\Delta\theta} > \widehat{\Delta\theta}$, which concludes the proof. ■

PROOF OF PROPOSITIONS 4 AND 5:

The mechanics of these proofs mimic that of Proposition 2 and 3. We skip the algebra and present the final expressions for the welfare difference between decentralized and centralized structures in polarized settings under perfect and private information. The welfare difference under perfect information is denoted by

$$\Phi(\Delta\theta, \Sigma\theta) = -\frac{1}{4(1+\lambda)^2} \left[\frac{\Delta\theta^2}{6} + \Sigma\theta^2 - \Sigma\theta\Delta\alpha - \lambda(2+\lambda)\Delta\alpha^2 \right].$$

It is straightforward to see that Φ is monotonically decreasing with respect to both $\Delta\theta$ and $\Sigma\theta$. Thus, for any $\Delta\alpha$ and $\Sigma\theta$, there exists $\widehat{\Delta\theta}$ such that $\Phi(\widehat{\Delta\theta}, \Sigma\theta) = 0$. For any $\Delta\theta > \widehat{\Delta\theta}$ ($\Sigma\theta > \widehat{\Sigma\theta}$), $\Phi(\Delta\theta, \Sigma\theta) < 0$ and centralization is welfare superior. Analogously, for any $\Delta\alpha$ and $\Delta\theta$, there exists $\widehat{\Sigma\theta}$ such that $\Phi(\Delta\theta, \widehat{\Sigma\theta}) = 0$. For any $\Sigma\theta > \widehat{\Sigma\theta}$, $\Phi(\Delta\theta, \Sigma\theta) < 0$ and centralization is welfare superior.

The welfare difference under private information is given by

$$\Lambda(\Delta\theta, \Sigma\theta) = -\frac{1}{16(1+\lambda)} \left[\frac{11}{3}\Sigma\theta^2 + \frac{1}{6}\Delta\theta^2 - 8\Sigma\theta\Delta\alpha - 4\lambda(2+\lambda)\Delta\alpha^2 \right].$$

Again it is straightforward to check that Λ is decreasing with respect to both $\Delta\theta$ and $\Sigma\theta$. Thus, for any $\Delta\alpha$ and $\Sigma\theta$, there exists $\widetilde{\Delta\theta}$ such that $\Lambda(\widetilde{\Delta\theta}, \Sigma\theta) = 0$. For any $\Delta\theta > \widetilde{\Delta\theta}$, $\Lambda(\Delta\theta, \Sigma\theta) < 0$ and centralization is welfare superior. Analogously, for any $\Delta\alpha$ and $\Delta\theta$, there exists $\widetilde{\Sigma\theta}$ such that $\Lambda(\Delta\theta, \widetilde{\Sigma\theta}) = 0$. For any $\Sigma\theta > \widetilde{\Sigma\theta}$, $\Lambda(\Delta\theta, \Sigma\theta) < 0$ and centralization is welfare superior. Additionally, it is simple to check that $\Phi > \Lambda$ for every combination of parameters. This implies that $\widetilde{\Delta\theta} > \widehat{\Delta\theta}$ and $\widetilde{\Sigma\theta} > \widehat{\Sigma\theta}$, which concludes the proof. ■

REFERENCES

- Austen-Smith, David.** 1995. "Campaign Contribution and Access." *American Political Science Review* 89 (3): 566–81.
- Austen-Smith, David, and John R. Wright.** 1992. "Competitive lobbying for a legislator's vote." *Social Choice and Welfare* 9 (3): 229–57.
- Bardhan, Pranab, and Dilip Mookherjee.** 2000. "Capture and Governance at Local and National Levels." *American Economic Review* 90 (2): 135–39.
- Bennedsen, Morten, and Sven E. Feldmann.** 2006. "Information lobbying and political contributions." *Journal of Public Economics* 90 (4–5): 631–56.
- Bernheim, B. Douglas, and Michael D. Whinston.** 1986. "Menu Auctions, Resource Allocation, and Economic Influence." *Quarterly Journal of Economics* 101 (1): 1–31.
- Besley, Timothy, and Stephen Coate.** 2003. "Centralized versus decentralized provision of local public goods: A political economy approach." *Journal of Public Economics* 87 (12): 2611–37.

- Bordignon, Massimon, Luca Colombo, and Umberto Galmarini.** 2008. "Fiscal federalism and lobbying." *Journal of Public Economics* 92 (12): 2288–2301.
- de Melo, Jaime, Arvind Panagariya, and Dani Rodrik.** 1993. "The new regionalism: A country perspective." In *New Dimensions in Regional Integration*, edited by Jaime de Melo and Arvind Panagariya, 159–201. Cambridge: Centre for Economic Policy Research.
- Grossman, Gene M., and Elhanan Helpman.** 1994. "Protection for Sale." *American Economic Review* 84 (4): 833–50.
- Grossman, Gene M., and Elhanan Helpman.** 1995. "The politics of free-trade agreements." *American Economic Review* 85 (4): 667–90.
- Grossman, Gene M., and Elhanan Helpman.** 2001. *Special Interest Politics*. Cambridge: MIT Press.
- Inman, Robert P., and Daniel L. Rubinfeld.** 1997. "Rethinking Federalism." *Journal of Economic Perspectives* 11 (4): 43–64.
- Krishna, Pravin.** 1998. "Regionalism and Multilateralism: A Political Economy Approach." *Quarterly Journal of Economics* 113 (1): 227–51.
- Lima, Rafael Costa, and Humberto Moreira.** 2014. "Information transmission and inefficient lobbying." *Games and Economic Behavior* 86: 282–307.
- Lockwood, Ben.** 2002. "Distributive politics and the costs of centralization." *Review of Economic Studies* 69 (2): 313–37.
- Loeper, Antoine.** 2011. "Coordination in heterogeneous federal systems." *Journal of Public Economics* 95 (7–8): 900–912.
- Maggi, Giovanni, and Andrés Rodríguez-Clare.** 2007. "A Political-Economy Theory of Trade Agreements." *American Economic Review* 97 (4): 1374–1406.
- Martimort, David, and Humberto Moreira.** 2010. "Common agency and public good provision under asymmetric information." *Theoretical Economics* 5 (2): 159–213.
- Martimort, David, and Aggey Semenov.** 2008. "Ideological Uncertainty and Lobbying Competition." *Journal of Public Economics* 92 (3–4): 456–81.
- Maskin, Eric, and Jean Tirole.** 1990. "The Principal-Agent Relationship with an Informed Principal: The Case of Private Values." *Econometrica* 58 (2): 379–409.
- Oates, Wallace E.** 1972. *Fiscal Federalism*. New York: Harcourt Brace Jovanovich.
- Ornelas, Emanuel.** 2005a. "Endogenous free trade agreements and the multilateral trading system." *Journal of International Economics* 67 (2): 471–97.
- Ornelas, Emanuel.** 2005b. "Rent Destruction and the Political Viability of Trade Agreements." *Quarterly Journal of Economics* 120 (4): 1475–1506.
- Potters, Jan, and Frans van Winden.** 1992. "Lobbying and asymmetric information." *Public Choice* 74: 269–92.
- Redoano, Michela, and Kimberly A. Scharf.** 2004. "The political economy of policy centralization: Direct versus indirect representative democracy." *Journal of Public Economics* 88 (3–4): 799–817.
- Richardson, Martin.** 1993. "Endogenous protection and trade diversion." *Journal of International Economics* 34 (3–4): 309–24.
- Riley, John G.** 1979. "Informational Equilibrium." *Econometrica* 47 (2): 331–59.
- Seabright, Paul.** 1996. "Accountability and decentralization in government: An incomplete contracts model." *European Economic Review* 40 (1): 61–89.
- Spence, Michael.** 1973. "Job Market Signaling." *Quarterly Journal of Economics* 87 (3): 355–74.